

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: EXCRETION IN PLANTS (SUB-STRAND 2.3)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This Final Explanation brings together everything you have learned in Lessons 1 to 5. Think about the trees and plants around our school: the Acacia and Euphorbia that ooze sticky sap or resin when cut, the droplets along the edges of grass and arrowroot (taro) leaves on a humid morning, and the old yellow leaves that fall. Answer each question in your own words in the space under it. Use the evidence you collected on the nature walks, in the microscope work, in your community interviews and portfolio, and in the class tables. Where a question asks for a Claim, Evidence and Reasoning, write all three parts. Use correct biology words such as excretion, exudation, guttation, transpiration, stomata, lenticels, hydathodes and tissue fall.

Part 1: Why plants do not need kidneys

Plants have no kidneys, bladders or sweating systems, yet they do not poison themselves with their own wastes. (a) What is excretion? (b) Explain why plants do not need complex excretory organs like the kidneys of humans. Use evidence from the metabolic rate chart and the mesophyll diagrams of CO₂ and O₂ that you studied, and from what you learned about the plant parts where wastes are kept.

(a) Excretion is the removal of metabolic wastes (and toxic substances or excesses) from the body of an organism. (b) Plants do not need kidneys for three reasons. First, the metabolic rate chart showed that plants have a much lower metabolic rate than animals, so they make much less waste. Second, plants recycle gases between respiration and photosynthesis. The mesophyll diagrams showed that in the mesophyll cells, the oxygen made in photosynthesis is used for respiration, and the carbon dioxide made in respiration is used for photosynthesis, so these gases do not build up as waste. Third, plants store insoluble, harmless wastes such as resin, latex, gum and crystals in tissues that are not vital, such as bark, vacuoles and old leaves. Because little waste is made and much of it is reused or stored safely, no complex excretory organ is needed.

Part 2: How wastes leave the plant

Describe the different ways that plants get rid of wastes. For each route, name the waste (or substance) that leaves and the part of the plant involved. Then use your observations from the nature walk and the bell jar

Plants remove wastes in several ways. (1) Diffusion of gases: carbon dioxide (the waste gas of respiration) and oxygen (the by-product of photosynthesis that is not used up) leave through stomata in leaves and lenticels on stems. (2) Transpiration: water vapour leaves, mainly through stomata. (3) Exudation: gums, resins and latex, which are stored waste products, ooze out of the bark and other tissues, for example when they are cut. (4) Guttation: liquid droplets containing dissolved substances leave through hydathodes at leaf edges in humid conditions. (5) Tissue fall: wastes stored in leaves, bark and fruits leave when these parts are shed. (a) The milky latex from a cut Euphorbia and the amber gum on Acacia are exudation: stored waste products that ooze out and then harden. (b) The droplets on grass and arrowroot leaves at dawn are guttation. In the bell jar experiment, droplets formed

experiment to explain what is happening to (a) the milky sap on a cut Euphorbia and the hardened gum on Acacia, and (b) the droplets at the edges of grass and arrowroot leaves in the early morning. How is guttation different from transpiration?	at the leaf margins of the seedling under the bell jar, where the air was very humid, but not on the control seedling, which supports the idea that guttation happens in humid conditions. Guttation is different from transpiration because guttation gives out liquid droplets with dissolved substances through hydathodes in humid conditions, while transpiration loses water as invisible vapour, as seen in the water vapour droplets collected in the plastic bag over a branch.
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Part 3: Where do insoluble wastes go?

Write a Claim, Evidence and Reasoning (CER) explanation to answer this question: How do plants get rid of wastes that cannot dissolve in water or evaporate? Use what you saw when you compared young green leaves with old yellow or brown leaves, and what you saw under the microscope, including how the crystals behaved when you added dilute acid. Explain why falling leaves can be called excretion.	Claim: Plants excrete insoluble wastes by depositing them in old or non-vital tissues, which are then shed. Evidence: When we examined leaves with hand lenses and under the light microscope, the old yellow or brown leaves had more calcium oxalate crystals than the young green leaves. The crystals did not dissolve when we added dilute ethanoic acid, which shows they are insoluble, as calcium oxalate is. We also saw resin or gum on the bark of Acacia and Euphorbia. Reasoning: Insoluble wastes cannot be carried out as vapour or droplets. By locking them in forms such as crystals, tannins and resins inside tissues the plant no longer needs, the plant keeps them away from its living, working cells. When the old leaves, bark or fruits fall, the stored wastes leave the plant with them. This is why tissue fall counts as excretion, and it shows that plants can clear their wastes without any kidneys.
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Part 4: Why plant wastes are valuable in Kenya

Many plant excretory products sell well in Kenyan markets. (a) Name at least four types of plant excretory products from your class table, and for each give an example of a plant or place in Kenya and one use. (b) Explain why something that is waste to the plant is valuable to people. (c) Explain what the Lesson 5 discussion taught you about the risks of some of these products.	(a) Latex: the milky latex of Euphorbia and of the papaya plant; papaya latex gives papain, and latex from rubber trees is used to make rubber for industry. Tannins: tannins from plant bark and tea leaves, such as the tea grown in Kericho, are used for tanning leather. Gums and resins: gum arabic from Acacia, collected in northern Kenya and Turkana, is used as an adhesive and food thickener, and resin is used for sealing. Alkaloids: quinine, from the bark of the cinchona tree, is used as medicine for malaria. Essential oils: oils from lemongrass or eucalyptus leaves are used in medicine and other products. (b) The plant stores or releases these chemicals as wastes, but they have useful chemical properties that people can use in medicine, industry and food, such as stickiness, sealing, tanning ability or medicinal effect. Communities collect and sell them, so they bring income. (c) Some alkaloids are harmful or habit-forming when abused. Their use must be controlled to prevent drug and substance abuse, even though many of the same kinds of chemicals are useful as medicines when used correctly.
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Part 5: Final explanation and your portfolio

Write your own final explanation of the driving question: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Your answer must use a Claim, Evidence and Reasoning structure. Use at least one piece of evidence from your community interview and photography portfolio or poster. Finish with the new question that your group has about the use of these products.

Claim: Trees and crops clear their wastes without kidneys because they make little waste, recycle much of it, store the rest harmlessly and release some through several surfaces; and some of these wastes are valuable because people can use their chemical properties. Evidence: The chart showed that plants have lower metabolic rates than animals, and the mesophyll diagrams showed CO₂ and O₂ being recycled between respiration and photosynthesis. We saw gases and water vapour leave through stomata and lenticels, droplets of guttation on grass and arrowroot leaves, gum and latex oozing from cut Acacia and Euphorbia, and old leaves with more crystals falling. In our community interviews and portfolio, people described using gum arabic, latex, tea, quinine and tannins for medicine, sealing, drinks and leather, and selling them for income. Reasoning: Because wastes are few, reused, stored in non-vital tissues and then released by diffusion, transpiration, guttation, exudation and tissue fall, plants do not need complex excretory organs. What the plant discards or stores is useful to humans, so products such as gum, latex, tannins and alkaloids are sold in Kenyan markets, though habit-forming alkaloids must be used under control. New question: How can sustainable harvesting of useful plant excretory products be balanced with environmental conservation?

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why plants lack excretory organs	Gives all three reasons (low metabolic rate, recycling of CO ₂ and O ₂ , storage of insoluble wastes) and links each to evidence from the chart or mesophyll diagrams.	Gives two or three reasons correctly, with limited reference to evidence.	Gives one reason or confuses excretion with other processes, with little evidence.
Describing the routes of waste removal	Correctly names and explains diffusion, transpiration, exudation, guttation and tissue fall, and clearly distinguishes guttation from transpiration using local observations.	Names most routes correctly and links them to the phenomenon, but the distinction between guttation and transpiration is partly unclear.	Names few routes or mixes them up (for example guttation and transpiration).
Use of Claim-Evidence-Reasoning with data	Claim is clear; evidence is specific (microscope, hand lens, bell jar, portfolio); reasoning links the evidence to the claim.	Claim and evidence are present, but the reasoning is brief or the link is not fully explained.	Claim is missing or unsupported, or evidence is general and not from the lessons.
Economic value and risks of plant excretory products	Gives at least four correct products with Kenyan examples and uses, explains why waste to the plant is valuable, and discusses alkaloid risks and substance abuse.	Gives two or three products with uses and a partial explanation of value or risk.	Lists one product or gives uses without explanation, with no mention of risks.

Scientific vocabulary and communication	Uses terms such as excretion, stomata, lenticels, hydathodes, exudation and guttation accurately in a clear, organised answer and ends with a relevant new question.	Uses most terms correctly and is generally clear.	Uses few scientific terms correctly and the answer is hard to follow.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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